Co2 Footprint Tracker



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PROJECT PROFILE

Project Name	Co2 Footprint Tracker
Project-Area	Lightweight Web Application
Objective	CO_2 Footprint Tracker web application is to empower individuals and organizations to understand, monitor, and reduce their carbon footprint. The application aims to achieve this by providing tools and insights to track emissions associated with daily activities, products, or services.
Technology	MS SQL Server, JSP, JAVA, spring Boot
Front-End	JSP ,HTML, CSS, TypeScript
Back-End	Java, Spring Boot, Web API, MS SQL Server Express

PROJECT DEFINITION

- The **CO_2 Footprint Tracker** is a web-based application designed to help users calculate, monitor, and reduce their carbon emissions. By leveraging user-provided data, the application provides personalized insights into CO_2 contributions from daily activities such as transportation, energy use, and consumption patterns.
- The project aims to promote environmental awareness and sustainable behavior through an interactive platform offering educational resources, actionable recommendations, and progress tracking.

PROPOSED SYSTEM

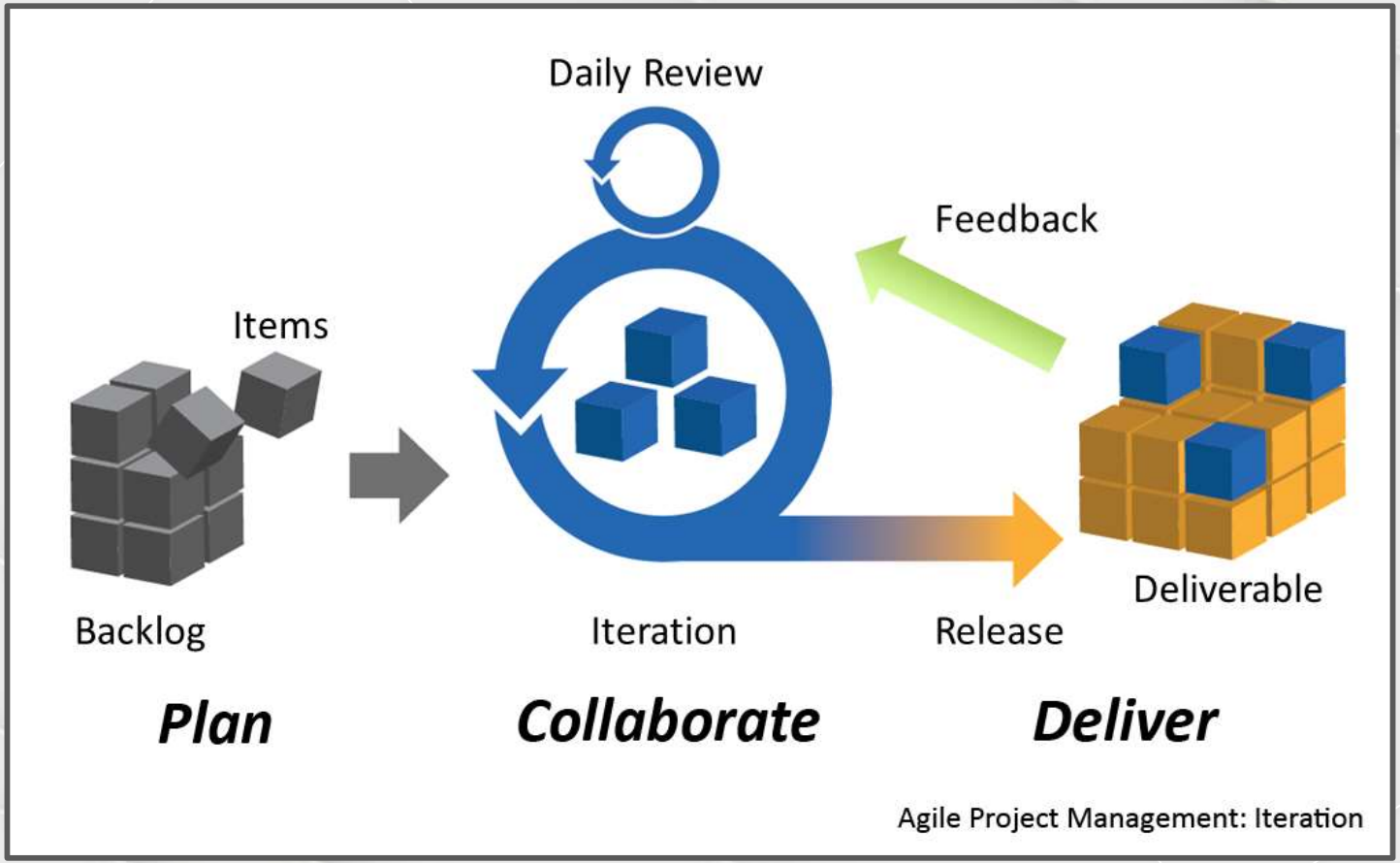
- Proposed System is the web-based project , which helps users to split amount between no. of members as well as keeps track of spent amount respectively.
- User can instantly split the spent amount between all group members by just clicking one button which might take hours to simplify expenditure.
- Here user don’t need to remember about its expenses and also to remind pending payments to his/her friends.
- It sends notification for settlements.
- Major advantage is ‘SIMPLIFY DEPT’ feature, which enables user to avoid multiple transaction.
- User gets better security. Application uses JWT-Bearer for security to stop intruders.
- This web-app supports most of the browsers, so there is no need of minimum hardware or software requirements.

TOOLS & TECHNOLOGY

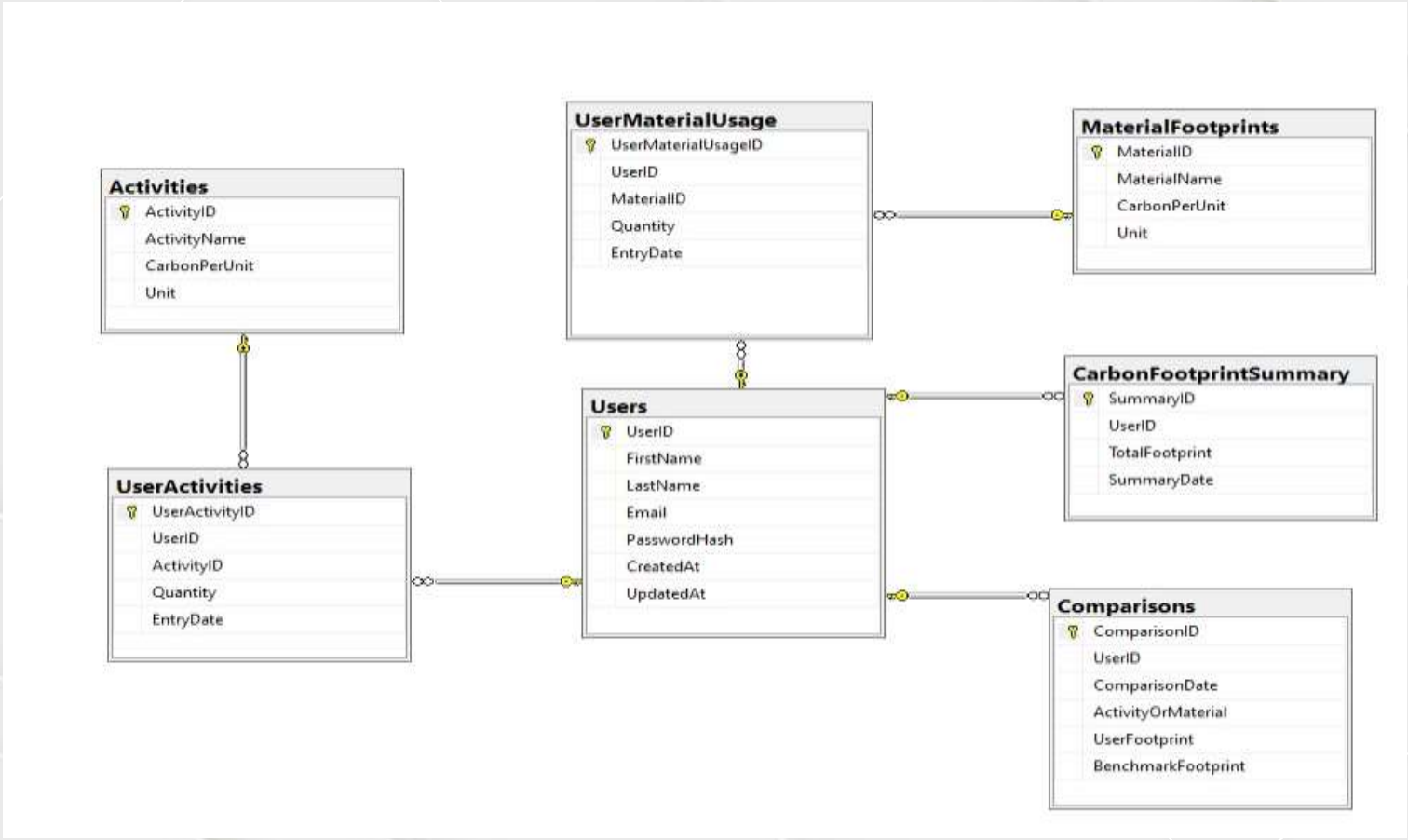
FRONT-END	
JSP	HTML
CSS	JAVA SCRIPT

BACK-END	
spring Boot	JAVA
MS SQL Server	Version 23

SYSTEM MODEL USED



ER DIAGRAM


Activities
ActivityID
ActivityName
CarbonPerUnit
Unit
UserMaterialUsage
UserMaterialUsageID
UserID
MaterialID
Quantity
EntryDate
MaterialFootprints
MaterialID
MaterialName
CarbonPerUnit
Unit
CarbonFootprintSummary
SummaryID
UserID
TotalFootprint
SummaryDate
Users
UserID
FirstName
LastName
Email
PasswordHash
CreatedAt
UpdatedAt
UserActivities
UserActivityID
UserID
ActivityID
Quantity
EntryDate
Comparisons
ComparisonID
UserID
ComparisonDate
ActivityOrMaterial
UserFootprint
BenchmarkFootprint

UML DIAGRAMS

DATA DICTIONARY

ACTIVITY TABLE

Column Name	DataType	Constraints
ActivityId	Integer	PK, auto_increment
ActivityName	nvarchar	not null
CarbonPerUnit	Integer	not null
Unit	Integer	not null

UserActivities TABLE

Column Name	DataType	Constraints
UserActivityId	Int	PK, auto_increment
UserId	Int	FK references
ActivityId	Int	FK references
Quantity	Int	not null
EntryDate	Datetime	not null

UserMaterialUsage TABLE

Column Name	DataType	Constraints
UserMaterialUsageId	int	PK, auto_increment
UserId	int	FK references
MaterialId	int	not null
Quantity	int	not null
Entrydate	datetime	null

Users TABLE

Column Name	DataType	Constraints
UserId	Int	PK, auto_increment
Firstname	Nvarchar	FK references
Lastname	Nvarchar	FK references
Email	Nvarchar	not null
PasswordHash	Nvarchar	not null
CreatedAt	DateTime	Null
UpdatedAt	DateTime	Null

MaterialFootprints TABLE

Column Name	DataType	Constraints
MaterialId	int	PK, auto_increment
MaterialName	NVARCHAR	not null
CarbonPerUnit	int	not null
Unit	int	not null

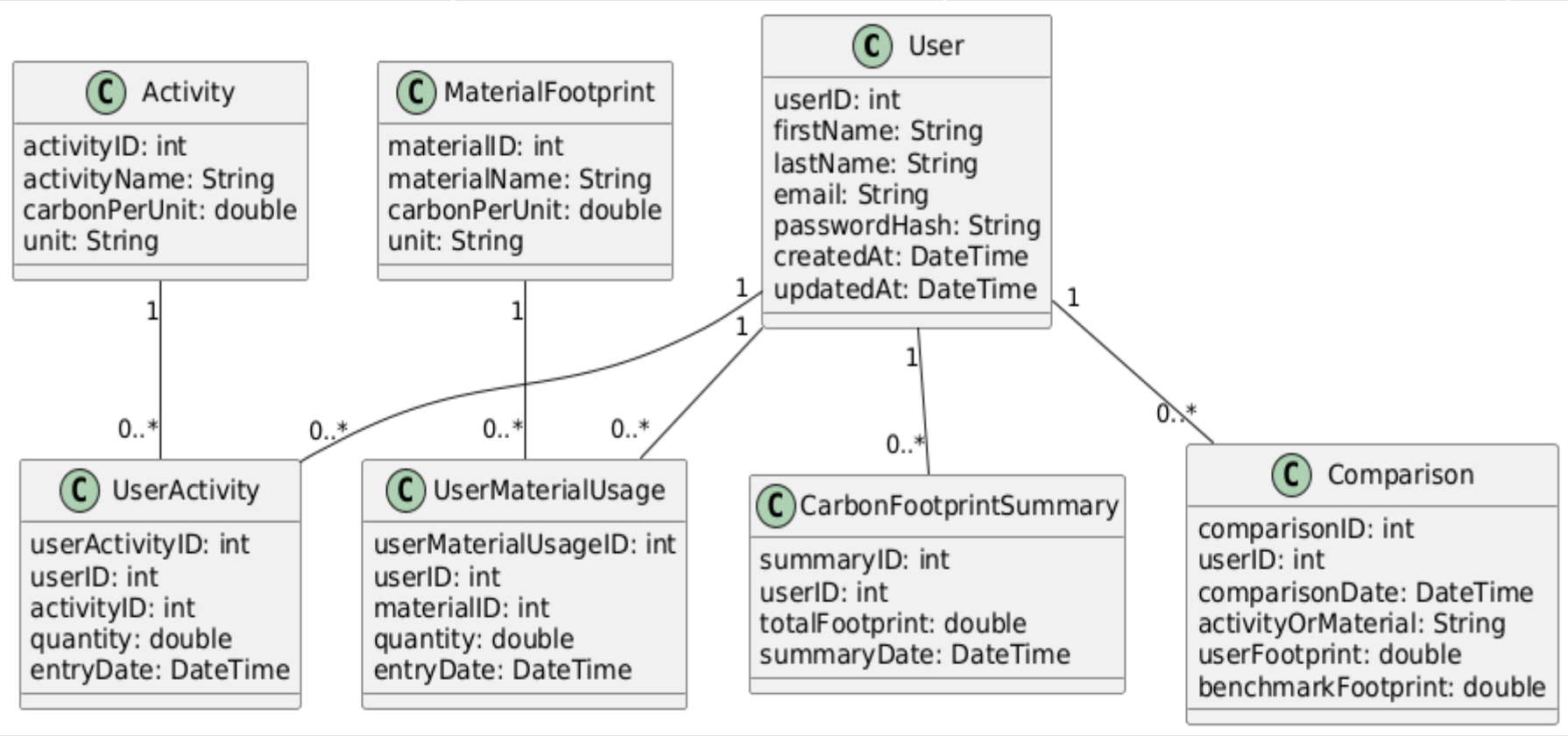
CarbonFootprintSummary TABLE

Column Name	DataType	Constraints
SummaryId	int	PK, auto_increment
userId	NVARCHAR	not null
TotalFootprint	int	FK references
SummaaryDate	DateTime	not null

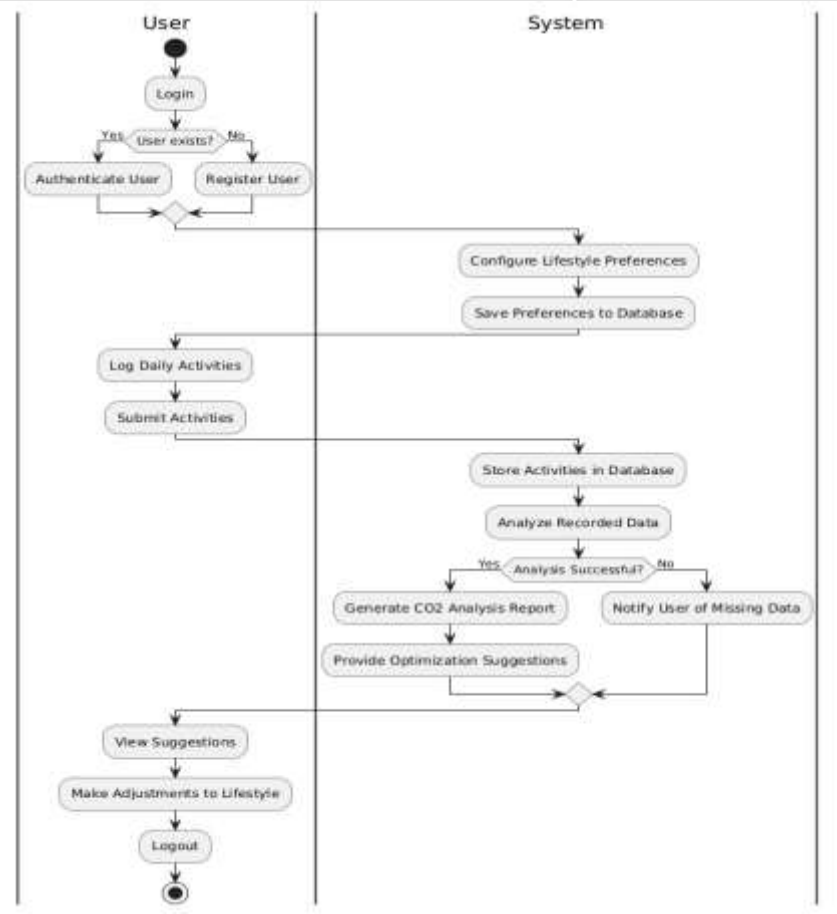
Comparisons TABLE

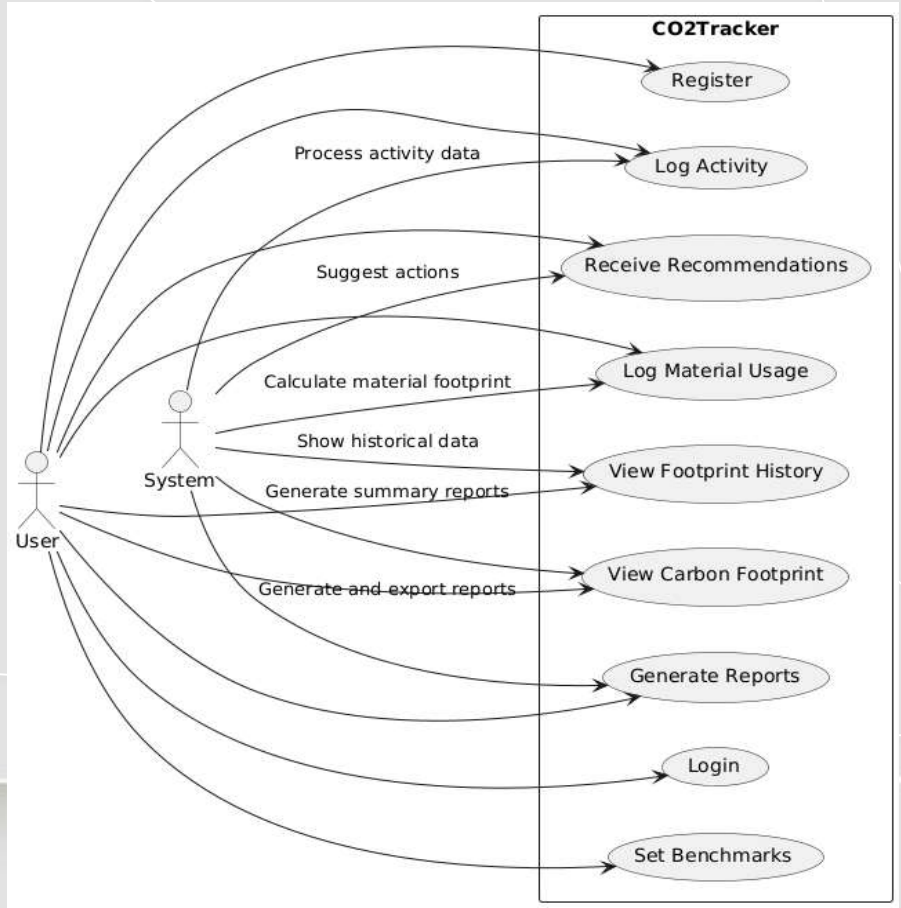
Column Name	DataType	Constraints
ComparisonsId	int	PK, auto_increment
UserId	NVARCHAR	not null
ComparisonsDate	DateTime	FK references
ActivityorMaterial	nvarchar	FK references
UserFootprint	int	not null
BenchmarkFootprint	int	not null

CLASS DIAGRAM

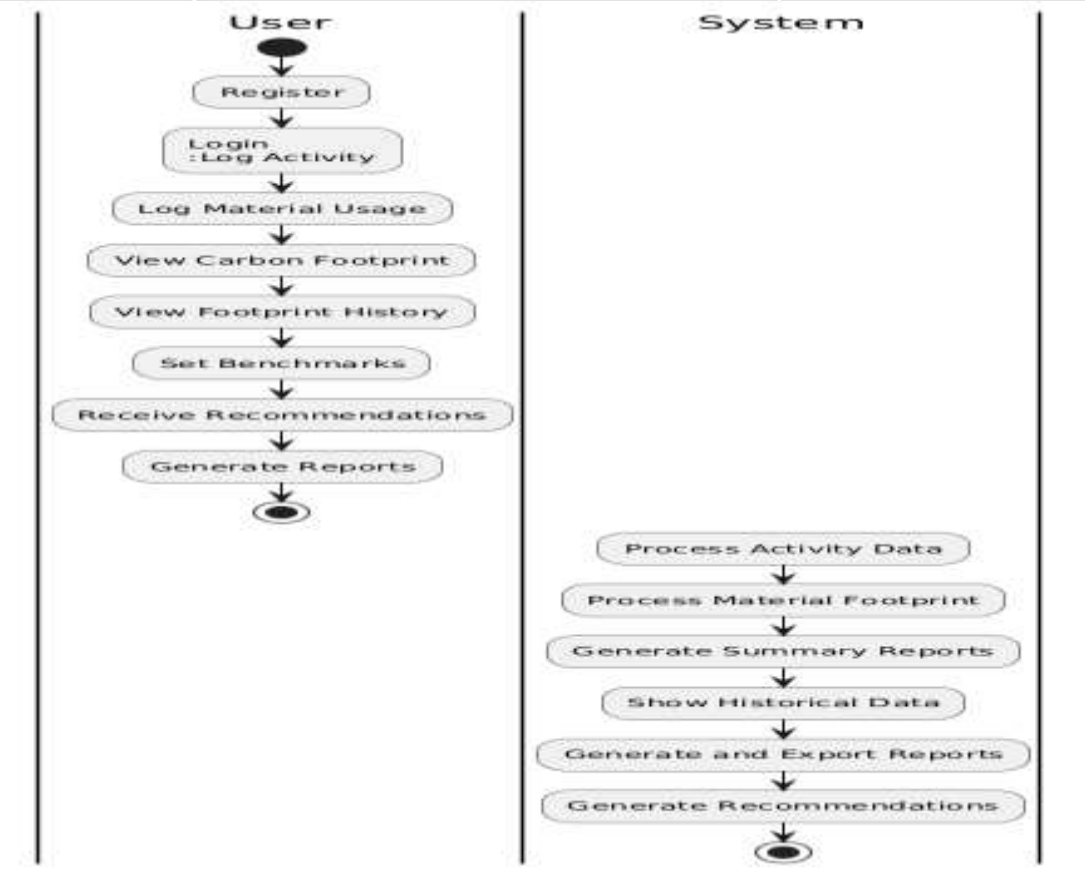


USE CASE DIAGRAM



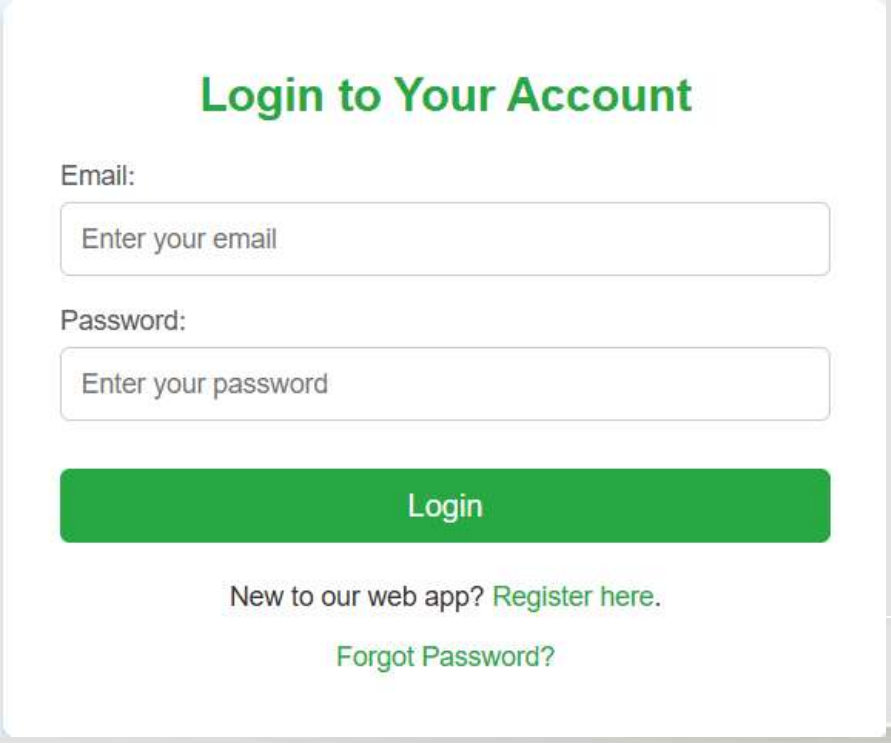


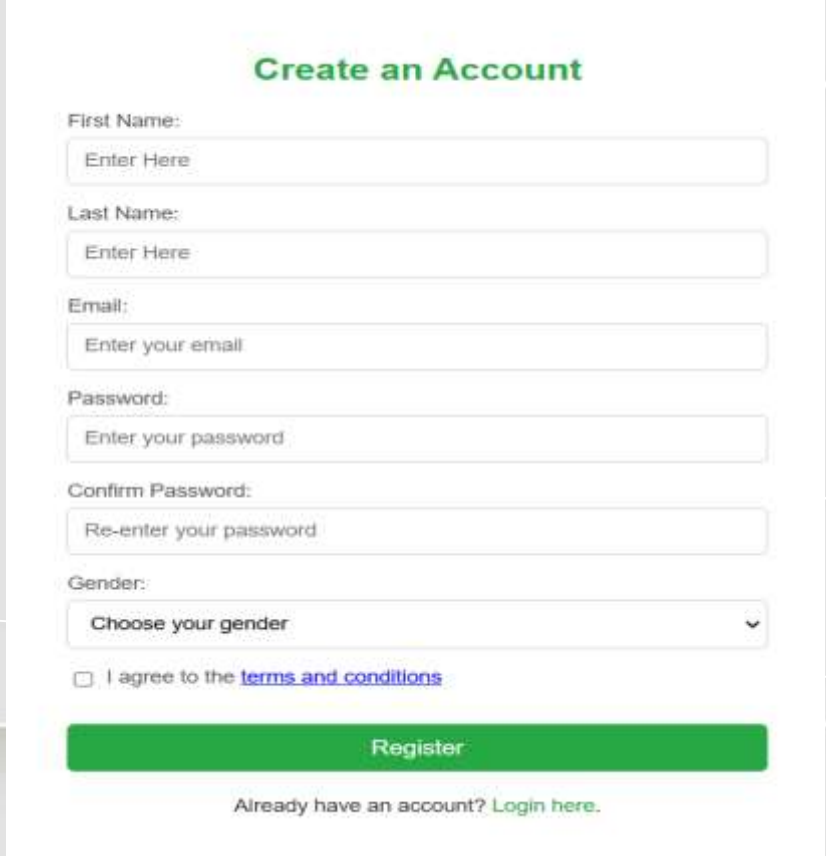
ACTIVITY DIAGRAM



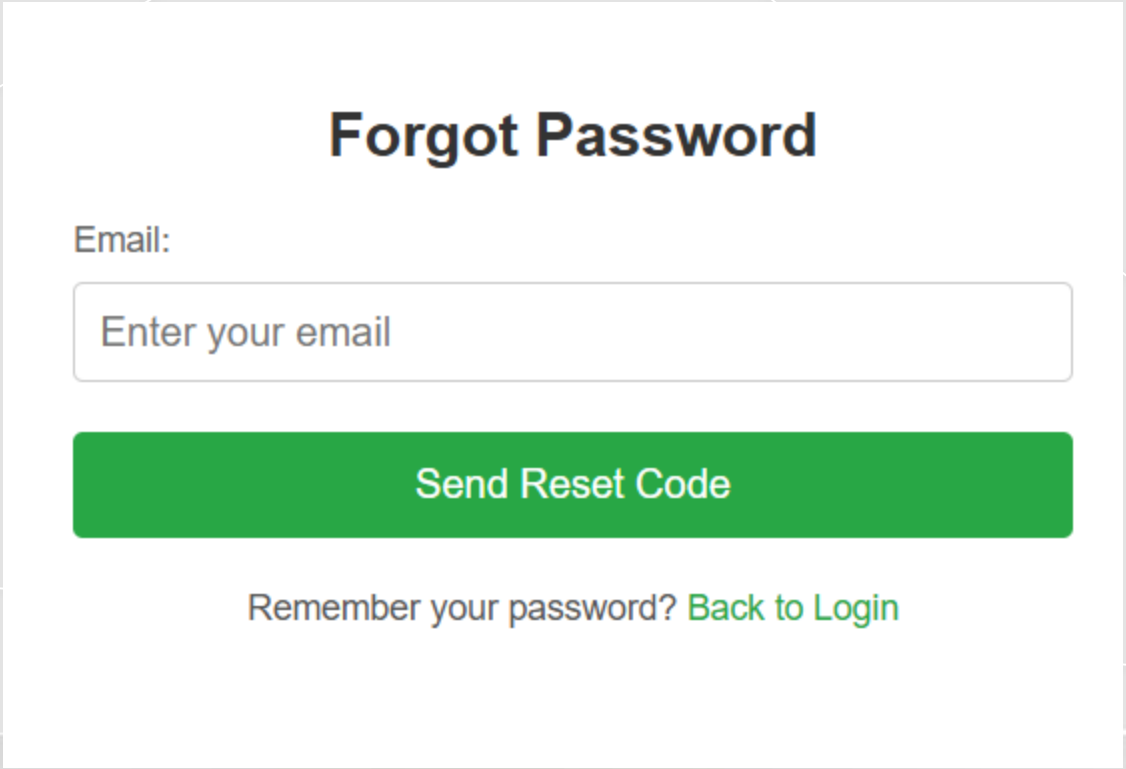
SCREENSHOTS

Login & Register

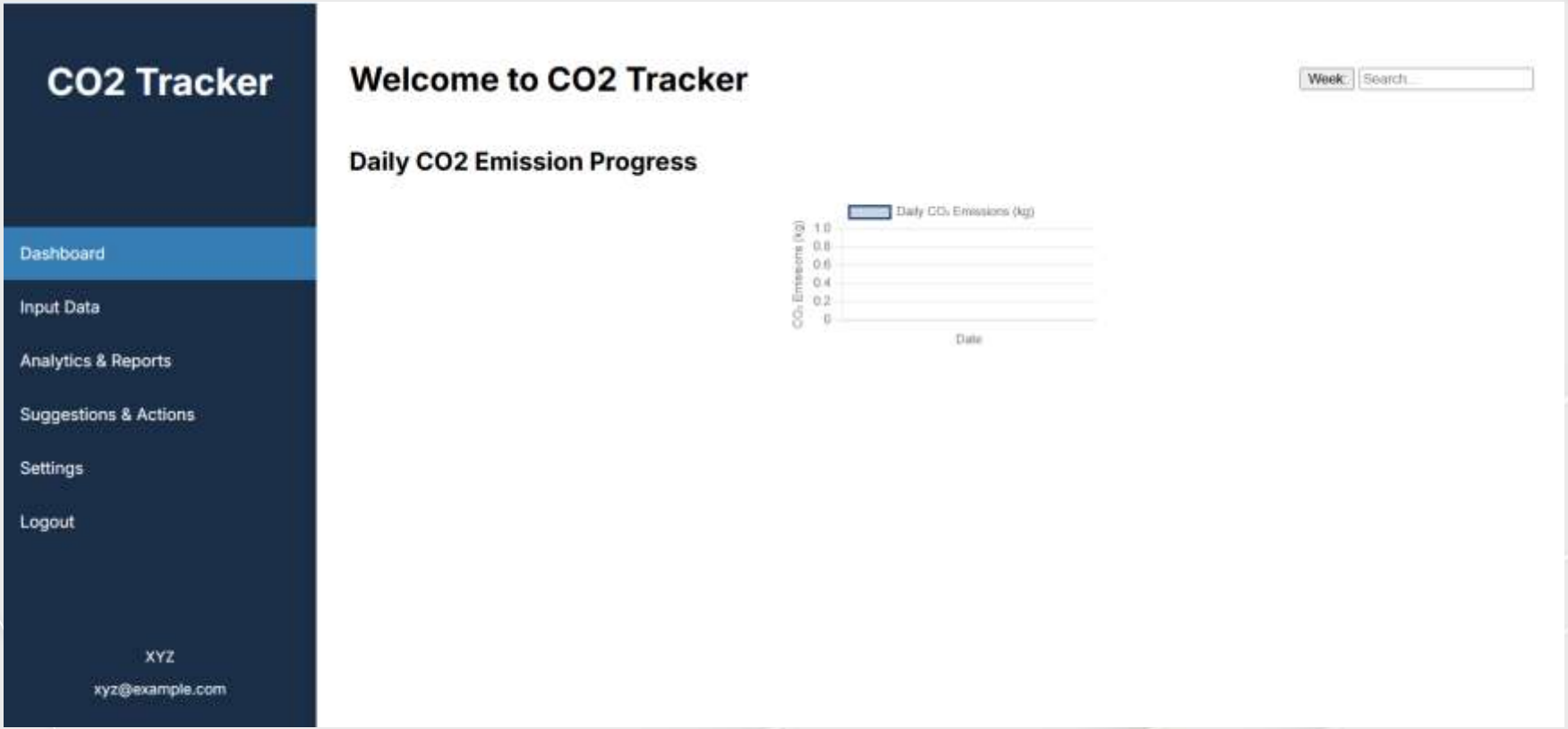




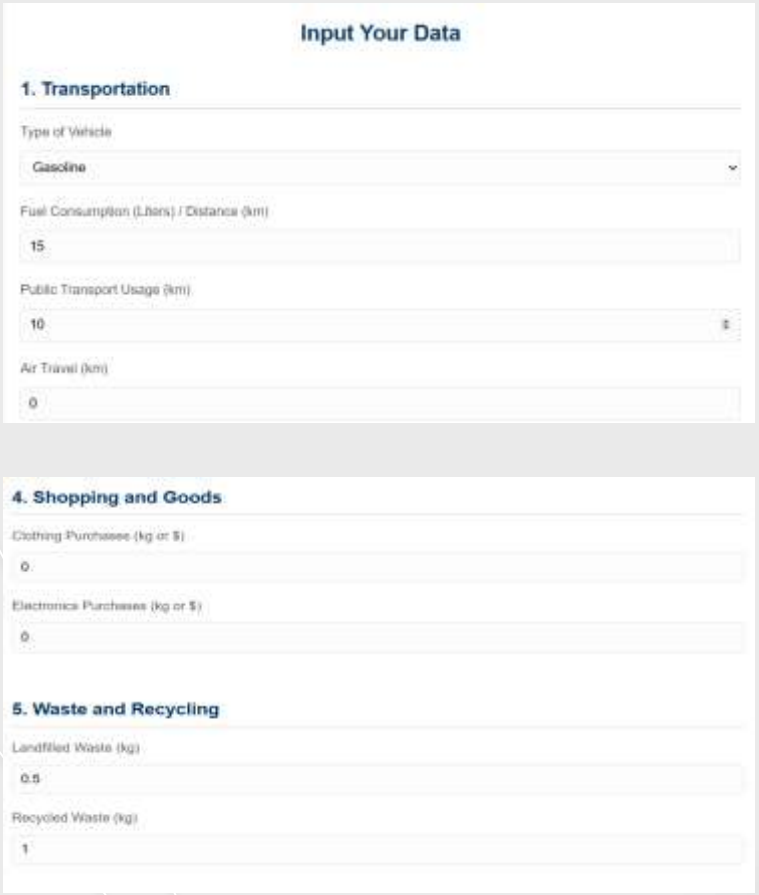
Forget Password



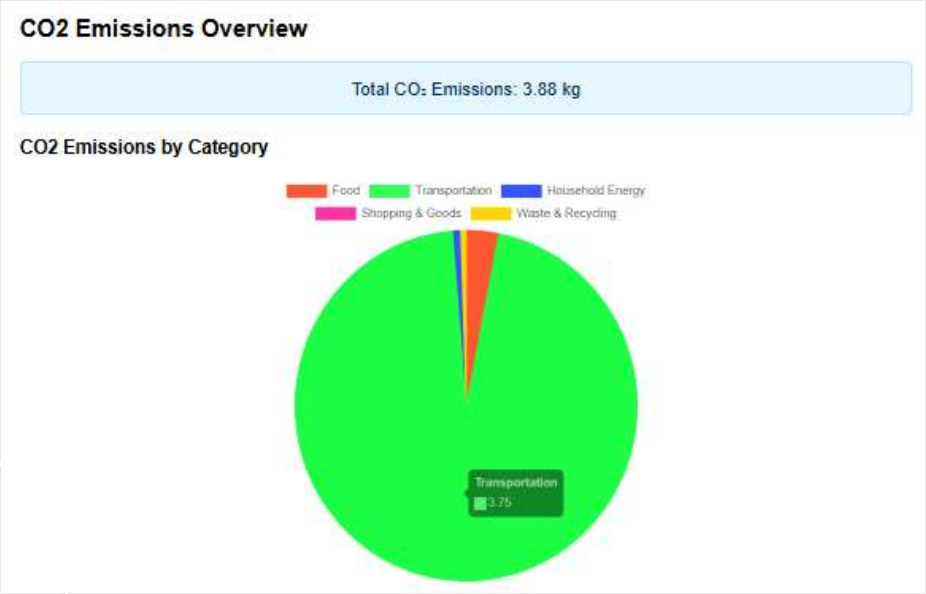
Dashboard Overview

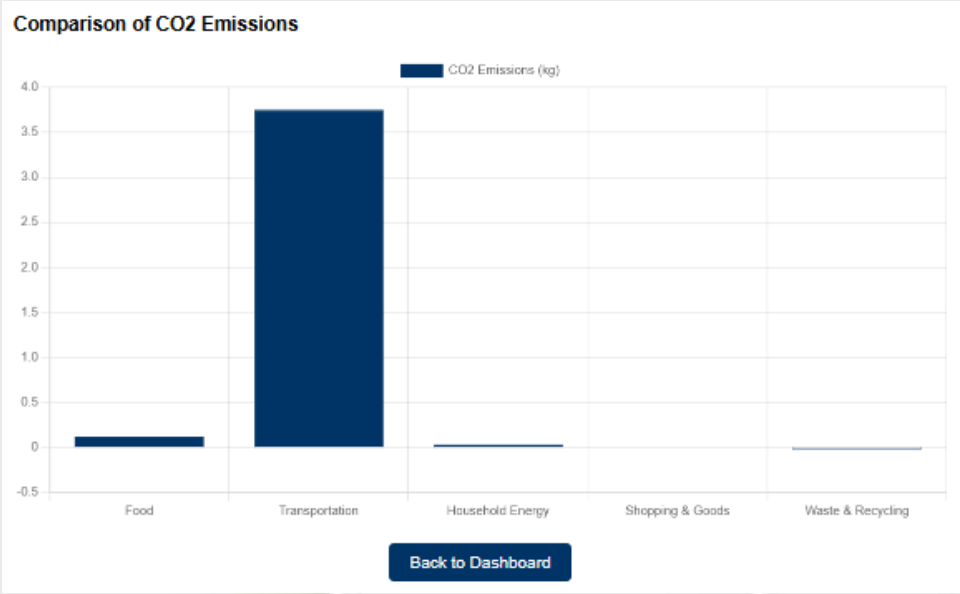


Input Data

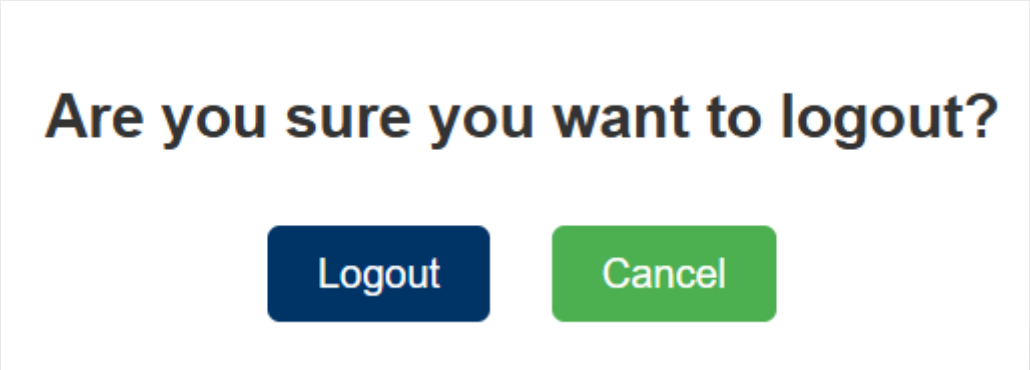

Input Your Data
1. Transportation
Type of Vehicle
Gasoline
Fuel Consumption (Liters) / Distance (km)
15
Public Transport Usage (km)
10
Air Travel (km)
0
4. Shopping and Goods
Clothing Purchases (kg or $)
0
Electronics Purchases (kg or $)
0
5. Waste and Recycling
Landfilled Waste (kg)
0.5
Recycled Waste (kg)
1
2. Household Energy Use
Electricity Usage (kWh)
20
Heating Type
Electricity
3. Food Consumption
Meat Consumption (kg)
1
Dairy Products (kg)
2
Plant-Based Foods (kg)
1

CO2 Tracker - Analytics & Reports





Logout



Thank you..